SOME BENEFITS OF ADVANCES IN COMPUTER CHESS

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1. INTRODUCTION

I am convinced that the psychological understanding of human top achievements will benefit substantially from intelligent analyses of what is going on in the contest between man and machine. It is an instructive test case regarding the question posed by the philosopher Dreyfus (1972): *What computers can't do* — and, it must be added: some humans can do. I shall not go into the details of what I have said about this subject on previous occasions but as an introduction to subsequent comments I refer to the contents of my Cologne paper on *Intuition in Chess* (De Groot, 1986).
In the Cologne paper, I sort of jumped from my analysis of intuition to the prediction that computer chess will not produce the World Champion until, say, after January 1, 2000. Before getting back to that prediction, however, I would like to report briefly about some developments that have happened meanwhile.

2. OPPOSITE VIEWS: A FAIR BET

Having received the 1986 ICCA article, Herbert Simon sent me a letter, the second sentence of which reads: "I think that I agree fully with your point of view about intuition, but not about your chess prediction". He then refers to a section in the second edition of *The Sciences of the Artificial* (Simon, 1970) where intuition is discussed and acknowledged as "a genuine enough phenomenon". However, the text then goes on to say that it "can be explained rather simply: most intuitive leaps are acts of recognition" — and this is illustrated by the game of chess; I shall not embark on the details of this argument by example. Its tenor can be assumed to be well-known. The concluding sentence of the section headed "Intuition" states that "recognition capabilities, and the information associated with the patterns that can be recognized, constitute a very large component of chess skill". In his letter, Simon adds that the evidence underlying his point of view has been provided by my work, his work, and that of a few others. "This evidence", he writes, "seems to me to demonstrate pretty convincingly that:

intuition = recognition, based on perception and providing access
to information stored in Long Term Memory".

In the next paragraph of his letter he then concludes: "If that's the correct interpretation, then contemporary chess programs, like Berliner's HITECH, have lots of intuition — though not enough yet to handle Grandmasters. But the year 2000 is still 14 years off, and at the rate things are progressing my own bet is that we'll reach the goal before then". It will be clear that we neatly agree to disagree here — not only on the specific prediction but also on the "lots of intuition" and the interpretation of the experimental and analytical evidence. In addition, I am afraid that Herbert Simon does not really "agree fully" with my point of view about intuition. Beyond that, there are fundamental, substantive differences between our views as well. I shall try to describe these briefly as I see them.

Logically, the locus of our disagreement can be specified quite simply. In my view, Simon is right in assuming that perceptual recognition is akin to intuition, or even a particular form of intuitive processing. I also endorse his statement that recognition capabilities constitute a large component of chess skill — "large" rather than "very large" for that matter — and that they do provide access to Long Term Memory. These points, however, solely support the statements that acts of recognition are important, and intuitive in nature. The contrapositive of the latter, namely that all intuitive operations are of the nature of recognition, is not supported by Simon's argument. I claim that there is much more to intuition than perceptually based recognition; Simon obviously does not endorse this claim. We also appear to have different conceptions regarding the contents of that Long Term Memory to which recognition acts give access. According to his hypothesis, intuition is over once LTM is reached; whereas in my view it is in LTM that, in addition to, and attached to all elements of *knowledge*, processing *skills* are elicited, many of which are intuitive heuristics.

Two other aspects are worth mentioning. First, while acknowledging holistic mental processes in the *concrete*, perceptual realm, Simon's explanation neglects all processing in which non-perceptual, *abstract mental inputs* are managed and integrated intuitively — whether along with concrete inputs or not. Take, for instance, the thinking subject's conception of the situation S, that is, "the present state of development of the total problem" (De Groot, 1978), on which the next operation, O, is to be based. Newell and Simon (1972) call this the subject's present "knowledge state"; but this simplifying label does not make it less abstract. It cannot be projected adequately onto a "permanent template of a chess-board" (Simon's terminology, in his letter), because it concerns a judgmental conception of the state of processing reached, that is, of the state of an *argumentation problem* (or process) rather than of a chess-playing problem. Its elements include intuitive *generalizations* of (in part intuitively completed) search outcomes, along with considerations of process management. In the latter respect, the human player is definitely not (solely) an automaton. He regularly readjusts his searching, evaluative, and decision-making strategies — in particular, consciously, in critical phases of his thought process. And he does so by abstract and intuitive reasoning [1].

This leads naturally to a second aspect of the difference between our two views. Simon's recognition hypothesis gives a solely *reproductive* explanation of a process that is *productive* in nature. Recognition acts, providing access to knowledge in memory, it is granted, are certainly omnipresent and indispensable in chess thinking; so are the manifold intuitive steps in productive reasoning, which I stress and Simon ignores.

In my answer to Herbert Simon, in this vein but worded differently, I stimulated him to write an article for publication in the ICCA Journal, possibly to be co-authored by Hans Berliner, in which he would take up and criticize my arguments and defend his counter-wager [2]. I did not get a written answer to the challenge; but a few weeks ago I happened to meet him in person . The available time was short; but in addition to some personal exchanges and shop talk on other matters, we discussed and attempted to clarify our respective positions regarding computer chess. In the end Simon promised to try to find the time — in his heavily-loaded work schedule — to write out his counter-bet argument to be submitted to the ICCA-Journal Editors. So there is some hope for a bet where two reputations are at stake — as they should be. Besides, as Simon remarked, his statistical expectancy of being alive in the year 2000 does not differ much from mine. In this "vital" respect, too, the bet can be said to be fair ...

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[1] Abstract reasoning can be observed in abundance in human protocols, in particular in the more problematic transitional phases of the thought process — provided the pertinent passages have not been cut out by the protocol analyst In such passages one finds, for example, (anticipated) generalizations of processing rules as well as of position characteristics, abstracted problem and solution attributes, weighing of operating strategies — risk taking? progressive deepening? control operations? stop rules? time expenditure? — and other considerations concerning the best process management It must be realized that the human decision maker aims not solely at finding the best move — the top of the iceberg — but has as his goal the construction to a subjectively convincing *argument* to support his move choice His calculations and almost continual evaluations serve a non-spatial, non-perceptual reasoning process (cf De Groot, 1978, Frijda and De Groot, 1981, Ch 7)

[2] About the same time and not yet informed about my proposal, the Editor-in-Chief happened to strenghten my plans by inviting Herbert Simon to write an article for the ICCA Journal

3. PROSPECTS OF OTHER STRATEGIES; WILL THE DATA BASE MAKE IT?

In extension of the previous arguments I should like to add some support to my reasons for not believing in what some term the unavoidable defeat of the World Champion within the next thirteen years. Although the reasons presented in the Cologne paper (De Groot, 1986) refer almost solely to improvements in selective search to the extent these depend on simulating, or at least borrowing from, human thinking, my guess about the future was not as wild as it might have appeared. In doing so, I perhaps skipped other possibilities of breakthroughs, in particular two types of them. Firstly, technical advances by which the operating efficiency of computers might be raised substantially — increased brute force for Shannon-A-type improvements — and secondly, the data-base approach.

Assuming, perhaps mistakenly, that further improvements in architecture and software techniques are relatively slow-moving and in the stage of diminishing returns, only *hardware breakthroughs* need to be considered. I am told that the recent research into materials that remain superconducting at the boiling point of liquid nitrogen may have revolutionary consequences for the speed limit in computer technology. My — uninformed — guess is, however, that even if such revolutionary inventions materialize, they are unlikely to be implemented within thirteen years from now to the effect of that unavoidable defeat of the human World Champion. If I am wrong I shall feel slain by brute force but not really unhappy. This type of possible developments is after all not what either Herbert Simon or I or anybody else in the betting game has in mind.

To some extent this also holds for the data-base approach as a particular way of incorporating knowledge into chess programs. Its idea is rather prosaic, perhaps somewhat threatening to chess-players, and not very attractive from a psychological point of view. Can it be successful, within the boundaries of current technology? It seems to me that one factor on which the feasibility of the data-base approach depends is that of the complexity of masterly chess. I define this complexity as the scope of the total problem space, in terms of the *number of possible master-game-likely positions*; or rather, in order to keep the pertinent numbers pronounceable, the 2-logarithm of that number, to be labeled the *variety of mgl-positions*. In the late sixties, and with a different aim, my former Ph.D. student R.W. Jongman and I carried out a few pertinent explorations. Regrettably, his thesis (Jongman, 1968) has never been translated, and my secondary analysis of his results in this respect has remained an unpublished chapter of a book — to be called "Perception and Memory in Chess" — yet to be completed. Still, some of our results and methods may be worth mentioning even now.

First, by plain counting and desk calculations the complexity of the problem space of *random positions* was found to be 143 as a 2-logarithm— i.e., their number is about 10^{43}, close to Shannon's (1950) estimate. For short, we shall write that the variety of these random positions is 143 bits.
Second, by introducing a few obvious legitimacy constraints — mainly regarding pawn structure — and a crude estimate of other ones, the variety of legal chess positions appeared to be of the order of — very roughly — 100 bits; a substantial reduction from the random, as could be expected.
Third, a sample of 192 positions after Black's 20th move, taken from then recent Russian tournament games, was used for estimating the pertinent variety of mgl-positions. By deriving uncertainty estimates from piece-location frequency distributions — rather than working less efficiently with square-occupancy frequencies — the sum of these 32 uncertainties provided a variety estimate of 70 for mgl-positions after Black's 20th move.
Fourth, since all such estimates tend to be inflated by the use of non-optimal codes and debatable for other reasons, a few more direct experimental methods were used — among these, variants of Shannon's (1951) guessing technique. The most interesting method, however, was based on the consideration that any mgl-position can be assumed to result from a series of "masterly moves", that is, moves a master would consider seriously as playable in the position in question. The idea is simple. Consider the total set of *mgl-positions* as the set of nodes in the theoretical tree consisting of all possible master level games up to their ends, i.e., up to resignable or clearly drawn positions. Then estimate their number on the basis of numbers of branchings — *mgl-playable alternatives* — per node, starting from the root. The point is that these

numbers of alternatives, and their distributions over categories of positions can be estimated very well — not by looking at played games but by simple interrogation experiments with masters as subjects. This is what Jongman has done, with one Grandmaster and two masters as subjects: Donner, Barendregt and Langeweg. With the experimenter, the three subjects were to reconsider, move by move, five recently played own games, up to and including Black's 20th move. In each of the positions where the subject had been on move — 20 per game except for a few earlier decisions — he was requested to judge as objectively as possible which alternatives, including the move actually played, he considered seriously playable. In the opening phase, double or multiple counts of alternatives resulting from transpositions in move sequence were eliminated as precisely as possible. The experimenter registered solely the numbers of alternatives.

In spite of the fact that the overall results confirmed an earlier estimate in my 1946 book based on the analysis of one (own) game, they were rather surprising. The average number of mgl-moves per positions — moves a master 'could have played' — turned out to be no higher than 1.76. Moreover, the distributions over frequencies of the three subjects were highly similar. Taken together, they judged only one move playable in no less than 54% of the positions. For two alternatives the percentage was 27; for three, 11; for four, 5; and for five, 2%. Six alternatives did not occur but there was one case — out of 288 — where seven alternatives were judged mgl-playable.

From the average number, 1.76, Jongman then derived a gross estimate of the *variety of mgl-positions after Black's* 20th move by simply taking the 2-logarithm of the fortieth power of this average — resulting in a variety of 33 bits. The upshot of his discussion of the reliability of this estimate is that the experimental outcomes are likely to somewhat underestimate the freedom of masterly choice, whereas his multiplicative procedure overestimates the variety of positions. By weighing these factors grossly — say, intuitively — he concluded that his estimate cannot be too far off the value of the statistic sought. As a conservative upper limit for the true value, he proposed 40 bits, an outcome he then used for his purposes, namely, to arrive at a quantitative assessment of the achievements of his (eight) master subjects in their main experimental task — consisting of reproducing unknown, briefly exposed chess positions after Black's 20th move.

Our present problem is of course a different one. It regards the feasibility of a successful data-base approach — in particular for the treatment of middle game and complex endgame positions. Even so, a few aspects of Jongman's work, whether still new after nineteen years or not, are of some interest.
Firstly, the general idea of a *hypothetical set of master-game-likely* positions, defined by the criterion of their resulting from a series of potentially masterly (mgl-)moves made by both players, seems well worth implementing. It is this universe the network of an actual data base should cover in the first place, if not solely. The general quantitative question of how large the holes in the network will remain (if any) depends on the variety of positions in the mgl universe.
Secondly, the principles of the experimental method employed by Jongman can provide sound estimates of a useful type of parameter — useful for estimating, level by level, the varieties of mgl-positions at the nth move. Furthermore, both the experimental and the subsequent variety-estimating procedures can be improved substantially by converting Jongman's own reservations — along with a few others he does not mention — to refinements that will enhance the reliability of the results [3].
Thirdly, the numerical results are striking. There is an enormous gap between the 33 bits result — or even the highly conservative upper limit of 40 bits — resulting from this experimental approach and the 70 bits variety resulting from other procedures not detailed here. It seems to me that the lower of the two figures is the more reliable one; that is, as a variety estimate regardless of coding problems. If this is so, then the evidence seems to suggest that collections of mgl-positions such as a data base would have to cover will also

[3] It would have been preferable, for instance (but much more time-consuming), also to consult one or more other master judges per position Another way of avoiding a too narrow or biased conception of mgl-playability — with the effect of somewhat increased freedom-of-choice figures — would be an instruction that emphasizes the testing-the-limits nature of the experiment As regards the variety estimate it would have been better, for example, to consider the 2-logarithms of the *products* of the experimentally *observed* freedom data per analyzed game as separate V(40)-estimates — eliminating the overestimation due to Jongman's arithmetic averaging Here, V(40) is to be read as the variety at ply 40, as it is in the remainder of this article

have much lower varieties than is generally assumed. Is this suggestion valid? It is, in fact, what I proposed in my unpublished book chapter — and believed until recently. The underlying reasoning is based on three considerations. Firstly, it is evident that the varieties of mgl-positions after n half-moves — and even before — fall short of the growth rate of Jongman's exponential function. Secondly, there must be an n-value at which the true variety as a function of n reaches a maximum. Thirdly, if this value of n is assumed not to lie too far above 40 ply — with its upper limit V(40) = 40 bits — then the total mgl-variety, summed over all values of n, should be of the order of no more than 50. However, I now think that this assumption of a rather early stop to growth is untenable.

I shall not attempt to solve this problem. But I would like to emphasize its relevance, firstly for the theory of the game of chess, secondly for predicting the feasibility of the data-base approach. As I see it, *the problem* of assessing the complexity of masterly chess, that is, the scope of its problem space, or, the total variety of mgl positions with which a data base approach will have to reckon, amounts to designing a sound *formula for V(n), the variety after n ply* (from the initial position). Jongman has offered us a basically sound method of estimating the constant growth factor, the averaged mgl-freedom of choice, f, and of developing V(n) as the simplest possible exponential function: $V(n) = f^n$. As a minimal — and hopefully sufficient — refinement, a growth retardation factor is called for, with a slowly increasing influence, to account for the collective effects of the increasing probability of move sequence transpositions, and the decreasing number of games still undecided. This "accounting for" is of course a very tricky problem. But I am convinced that a reasonably valid solution is possible — provided that the V-function is theoretically tenable and its parameters are estimated and tested, not on the sole basis of statistical data, but also on that of a few other, sophisticated experimental studies [4].

4. BACK TO THE PREDICTION PROBLEM

The exploration of the complexity of masterly chess has provided us with not unreasonable variety estimates for values of n up to 40 but not given us much of a hold on what happens to V(n) beyond that number of half-moves. The problem of the "growth retardation factor" has remained unsolved. Its trickiness itself, however, that is, the difficulty of the sole task of delimiting the scope of the problem space a data-base approach should cover has strengthened — perhaps irrationally — my intuitive conviction that Grandmasters have no reason, as yet, to be afraid of it; that is, before the year 2000. Another conclusion I draw from the exploration is that even data-base problems can be fun — and of interest, if not for psychology, then at least for the applied theory of games (cf. Dekker et al., 1987).

For reasons of space, I omit again a discussion on *selective search*, to the extent that it is based on the introduction of more 'human' ways of processing. I still believe that, in the long run, substantial advances in this respect can be expected. Combined strategies are also possible, for instance using a data base as a plausible move generator along with a subsequent arguing and decision-making process that simulates human reasoning in the vein of progressive deepening. However, any such set-up involves the handling of uncertainties, the production of anticipations, or, in brief: the art of fuzzy, intuitive processing. It is true that intuitive ways of thinking may be converted in part to formal representations — and thereby be

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[4] *Game-length frequency statistics* are obviously important, as inputs to the assessment of the retardation factor If the empirical distribution shows that a certain level of n, say of n = 60 or 70, half of the master games have ended, the Jongman estimate of V(n) is to be decreased by one bit, and so on However, this treatment rests on the assumption that all resigned and drawn games were no longer worth continuing, i e , that the end position itself and all subsequent positions that could have resulted from continued mgl-play do not belong to the set But this is dubious for draws by agreement On the other hand, obviously lost or drawn games are sometimes continued Such cases demand a special treatment for which masterly expertise is indispensable Much harder is the problem of curbing the influence of *multiple counts* of the same position Endgame positions in particular could have originated not only from a series of move-sequence transpositions but also from completely different games, and at different levels of n Except for introducing a rather arbitrary growth-retarding "multiple count factor" and testing it on other criteria of plausibility of outcomes, I have no specific suggestion to offer regarding its treatment In any case, here again, expert judgment with masters as subjects — e g , in retrogade-analysis experiments using positions with medium values of n— could provide very helpful inputs

falsified as being in the realm of intuition — but this is a very hard job. Heuristics research necessarily progresses too slowly to enable programmers to beat the World Champion before 2000. There are other arguments: for instance, that human cunning in exploiting the weakness of computer programs must not be underrated. Beating a human champion who avoids combinatorial risks and patiently awaits his opportunity to break a series of draws by a positional victory is a very hard job. In conclusion, I maintain that within thirteen years it cannot be done, but also that it must be attempted. In this area the goal of designing a machine that will beat the best of men is worth our best effort. Working for it is not only fun but also of scientific, cultural and philosophical interest.

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A.D. DE GROOT AT NOORDWIJKERHOUT

I intuit,
bytes can't do it.